

Dilawar

Data Analyst | Machine Learning | CS Student

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Education

Sukkur IBA University 2025 – Present

BS Computer Science — 2nd Semester Sukkur, Pakistan

Relevant Coursework: Data Structures, Object-Oriented Programming, Database Systems, Calculus

Skills

Languages: Python, SQL, Java, C++

Analytics: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI, Excel, EDA, Statistical Analysis

Machine Learning: Scikit-learn, Random Forest, Logistic Regression, SVM, LSTM, K-Means, SMOTE, XGBoost

Databases: MySQL, JDBC, Relational Design (3NF), Query Optimisation

Tools: Git & GitHub, Streamlit, Jupyter Notebook, IntelliJ IDEA, Kaggle

Certifications

Google Advanced Data Analytics Professional Certificate

Coursera / Google — 7-course program

Python, Statistics, Machine Learning, Tableau, Project Planning, Data Communication

Flutter Development Workshop

Sukkur IBA University

Data Analytics & Machine Learning Projects

Credit Card Fraud Detection System

Python | Random Forest | SMOTE | Streamlit | 284,807 transactions | streamlit.app/fraud-detection | GitHub

Built an end-to-end fraud detection pipeline on 284,807 real European transactions with a 577:1 class imbalance — the most challenging real-world ML scenario.

Applied SMOTE to handle severe imbalance; trained and compared Logistic Regression, Random Forest, and XGBoost — achieving 85% F1 Score and 97% ROC-AUC.

Deployed a live Streamlit web app with real-time fraud prediction, 4 interactive tabs, 8 business charts, and feature importance analysis — accessible via public URL.

Demonstrated why accuracy is misleading on imbalanced data and used Precision, Recall, and F1 as the correct evaluation metrics.

Pakistan Automotive Industry Analysis (PAMA)

Python | Pandas | Matplotlib | Streamlit | 17,723 records | 2007–2024 | Live App | GitHub

Analysed 17,723 records of real Pakistani automotive data (PAMA) spanning 2007–2024 — production and sales across cars, motorcycles, trucks, and emerging electric vehicles.

Uncovered that motorcycles dominate with 73.2% average market share, Suzuki leads car sales, and cycles show zero local production across 17 years — fully import-dependent.

Identified monsoon-driven seasonal dips in July & September, May as the peak sales month, and quantified economic shocks including COVID-19, IMF austerity (2018–2020), and 2022 floods.

Deployed a 4-tab interactive Streamlit dashboard with year/category filters, YoY growth charts, brand competition analysis, and a detailed business insights panel.

Brazilian E-Commerce Sales Analytics (Olist)

Python | SQL (SQLite) | Pandas | Matplotlib | Streamlit | 96,426 orders | Live App | GitHub

Merged 9 relational CSV files into a master table using Python and SQLite SQL queries — joining orders, customers, products, sellers, payments, and reviews into a single analytical dataset.

Executed 5 SQL queries (GROUP BY, CASE WHEN, HAVING, ORDER BY) to identify top revenue categories, underperforming states, seasonal patterns, and delivery-satisfaction correlation.

Key findings: bed_bath_table leads revenue at R.6M, 0-7 day deliveries score 4.33/5 vs 3.01/5 for 22+ days, and remote states show 17-25 day delivery times suppressing demand.

Delivered a 5-tab Streamlit dashboard with live filters, business insights panel, and a formal business memo with 5 actionable recommendations for operations leadership.

Employee Attrition Prediction

Python | Scikit-learn | Random Forest | SMOTE | EDA | GitHub

Built an end-to-end ML pipeline to predict employee turnover; achieved 87% accuracy using Random Forest, outperforming baseline logistic regression by 12 percentage points.

Applied SMOTE and feature importance analysis to surface key HR risk factors for actionable retention decisions.

Customer Segmentation

Python | K-Means | Hierarchical Clustering | Seaborn | GitHub

Segmented customer base using K-Means and hierarchical clustering; determined optimal segment count via Elbow Method and Silhouette Score.

Produced detailed visual customer profiles enabling data-driven targeted marketing strategies.

Stock Exchange Price Prediction

Python | LSTM | Linear Regression | Time-Series | GitHub

Developed LSTM and Linear Regression models for stock price forecasting; engineered lag features and rolling statistics validated with RMSE and MAE metrics.

Titanic Survival Prediction

Python | Classification | Feature Engineering | Scikit-learn | GitHub

Compared Decision Tree, SVM, and Random Forest classifiers; achieved 82% test accuracy with full EDA, missing-value imputation, and feature engineering.

Software Development Projects

Online Tuition Management System

Java | Java Swing | JDBC | MySQL | 3NF Database | GitHub

Designed a full-stack desktop application with 7 modules backed by a normalised MySQL database with FK constraints, CRUD operations, and SQL injection prevention.

Quiz Application

Java | Java Swing | OOP | GitHub

Built an interactive quiz app with dynamic question loading, countdown timer, real-time score tracking, and replay functionality.

Activities

Kaggle Competitions — participated in multiple ML challenges; practised EDA, feature engineering, and competitive model building on real-world datasets.

Open Source — all projects publicly available on GitHub with documented code, charts, and structured READMEs.